



TECHNICAL NOTE

TN104

IRS COMMAND INTERFACE

REV: 3.0

REVISION HISTORY

Rev	Date	Change Description	Prepared By	Verified By	Approved By
1.0	16/01/2023	Applicable to Z175/ID09	M. Rispoli	L.	

CHANGE LOG

- Compressor: Added command to set the minimum selectable target compression;
- Config: Added command to change the compressor release behavior
- System: Added a command to test the SSR relay integrity;
- Biopsy: Added command to enable/disable the use of the Y upright position sensor;

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1. Document Overview

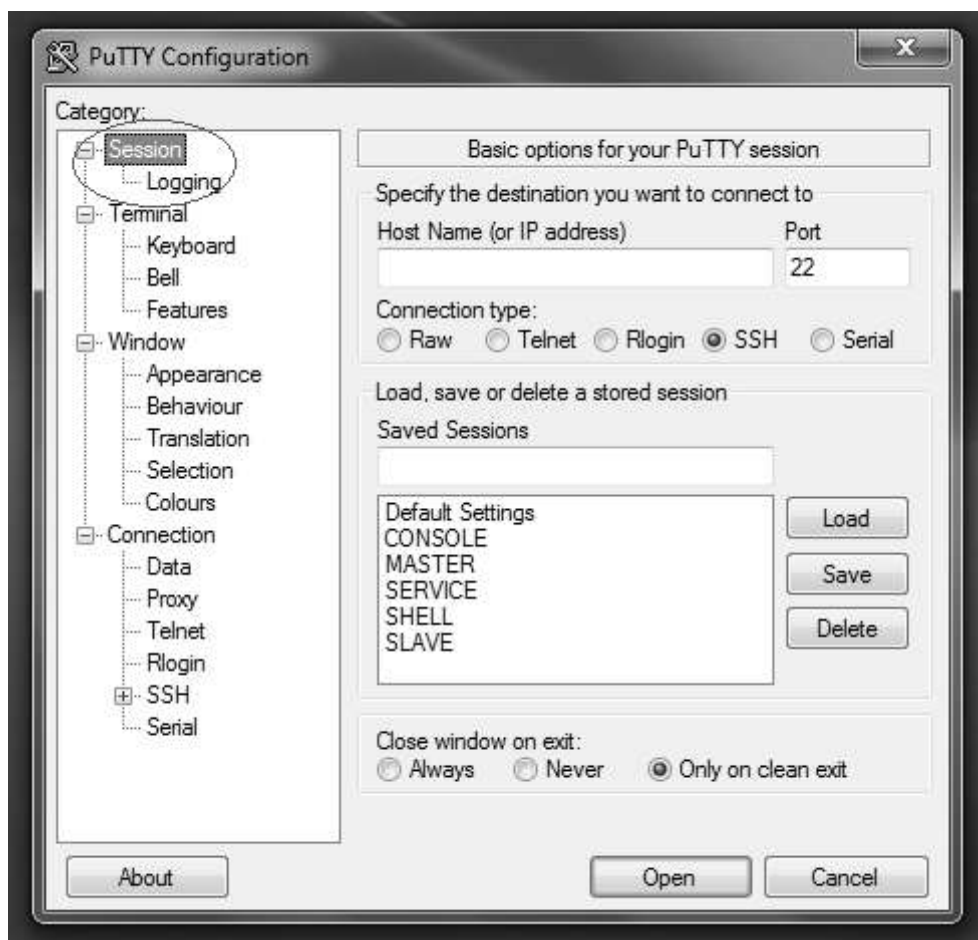
This document describes the functions of the DBTController IRS command interface.

2. IRS Overview

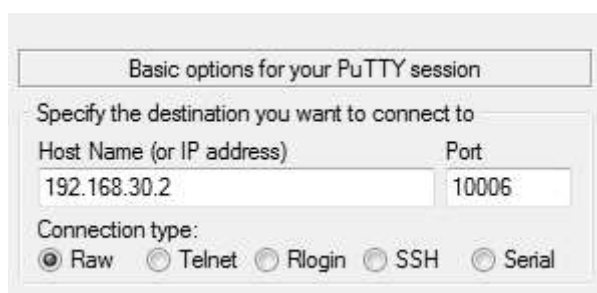
The DBTController IRS (Remote Service Interface) is basically a low level command "shell" providing textual commands that interact with the DBT with a very low level functions.

3. Activation

The DBTController IRS can be activated locally on the AWS station, launching the pre installed



3. Set the "Basic options.." fields as follows:



4. Press the button OPEN to start the session: If the connection success the a shell pop-up displaying a welcome row reporting the current revision version of the IRS interface.

6. Command syntax

The command syntax is easy:

[group name:] <command name> [parameter1, parameter2,...parameter-n]

Group Name Description

The **group_name** is a sub-set of commands grouped together

For example:

```
system:
system >
```

In the example above, the "system" sub group is entered, having access to a set of commands belonging to this group.

In order to get the command content of the group, it can be send the ? command into the group:

```
system: ?
```

NOTE: when a group is entered it is not necessary

In order to get the whole group availables, the user can type the command ? in the root of the Terminal:

Type the following commands:

```
..
?
```

The ".." command causes the Terminal enter the MAIN group (the root of all groups);

The command "?" request for the list of all available groups

7. Group command description

7.1. SYSTEM COMMANDS (system:)

7.1.1. **setDATE**

COMMAND

setDATE Year Month Day hour minute second

DESCRIPTION

Set the current display date

7.1.2. **getRevisions**

COMMAND

getRevisions

DESCRIPTION

Get the list of the firmware revisions in the package

7.1.3. **setShotMode**

COMMAND

atMode

N

tion (with detector)

Electrically Opened

COM

setCalibKv

DESCRIPTION

~~Set the kV calibration~~

COMMAND

Set the homo calibration Mode

NOTE: *the study is automatically Opened*

7.1.8. **setOperatingMode**

COMMAND

setOperatingMode

DESCRIPTION

Set the Operating mode

NOTE: *the study is NOT automatically Opened*

7.1.9. **reboot**

COMMAND

reboot

DESCRIPTION

The Master and Slave Terminals will reboot

7.1.10. **reboot**

COMMAND

setPark

DESCRIPTION

orrectly working."

If the SSR relay should be damaged, the command answers: " SSR seams to be broken."

NOTE: this command shall be used only with the system powered off for almost 1 minutes before to be repowered.

7.2. CONFIGURATION COMMANDS (config:)

7.2.1. **setAutoFilter**

COMMAND

setAutoFilter ON/OFF

DESCRIPTION

Activate/Deactivate the system option Auto Filter

If set to ON the system will activate the Filter selection.

If set to OFF the system will use the Fixed Filter (no filter selection).

NOTE: reboot the Gantry after modification

7.2.2. **setHsStarter**

COMMAND

setHsStarter ON/OFF

DESCRIPTION

Activate/Deactivate the High Speed Starter system option

N

7.2.3. **setArmMotor**

COMMAND

setArmMotor ON/OFF

DESCRIPTION

Activate/Deactivate the Motorized ARM

If set to ON the system will activate the ARM with Motor.

If set to OFF the system will provide a Manual ARM activation

t the Gantry after modification

7.2.4. **setTrxMotor**

COMMAND

setTrxMotor ON/OFF

DESCRIPTION

Activate/Deactivate the Tilting

NOTE: reboot the Gantry after modification

7.2.5. **setDemoMode**

COMMAND

setDemoMode ON/OFF

DESCRIPTION

Activate/Deactivate the Demo Mode

***NOTE: after the command execution Gantry automatically reboot ***

7.2.6. **enableAccessory**

COMMAND

enableAccessory ON/OFF

DESCRIPTION

Activate/Deactivate the test on the Protection Patient

7.2.7. **setSN**

COMMAND

setSN serial_number

COMMAND

DESCRIPTION

Set the Gantry serial number

NOTE: The `serial_number` shall be composed of only digits

Usage:

| `setSN 123456`

set the Gantry serial number to 123456

| `setSN .`

erases the serial number!

7.2.8. ***setPSW***

COMMAND

setPSW password_number

DESCRIPTION

Set the Gantry service password number

NOTE: The `password_number` shall be composed of only digits

7.2.9. ***enableStarterKeep***

COMMAND

enableStarterKeep ON/OFF

DESCRIPTION

Enables/Disables the Starter keeping mode

NOTE: For Low speed starter set this parameter to OFF!!

7.2.10. ***setTubeTemp***

COMMAND

setTubeTemp LT HT

DESCRIPTION

Set the Tube alarm threshold(HT) and the reset alarm threshold (LT)

- HT: is the Tube temperature in (°C) to trigger the Alarm;
- LT: is the Tube temperature in (°C) to reset the Alarm;

7.2.11. **setLanguage**

COMMAND

setLanguage [ITA/ENG/SPA/FRA/POR/RUS]

DESCRIPTION

Set the GUI current language

NOTE: The Language is immediatelly changed!!

7.2.12. **resetGantry**

COMMAND

resetGantry

DESCRIPTION

Clear the Gantry System Configuration file

Use this command when the Gantry shall be reconfigured for the hardware setup:

- Motor/Manual rotation;
- HS or LS starter;
- Tilting presence;

7.2.13. **resetKvCalib**

COMMAND

resetKvCalib

DESCRIPTION

Clear the kV-sense calibration file

Use this command when the kV read back calibration shall be removed.

NOTE: In case of switch between 35kV generator to 49kV or viceversa it is recommended to clear the calibration file!!

7.2.14. **sysBackup**

COMMAND

sysBackup backup_name

DESCRIPTION

Create a backup of both Master and Slave terminals

~~The command doesn't update the Devices. Use the download command (see below)~~

When the Collimation type is AUTOMATICA then the Gantry controls the collimation.
When the Collimation type is MANUALE then the collimation is controlled manually by IRS commands.

7.3.3.2. **setManual**

COMMAND
setManual
DESCRIPTION
Set Only the collimation format in Manual mode

The Filter setting remains in Automatic mode!!

7.3.3.3. **setAuto**

COMMAND
setAuto
DESCRIPTION
Set both the Filter and the Collimation format in Auto mode

7.3.3.4. **setL**

COMMAND
setL val
DESCRIPTION
Set the left manual blade position

*** This position is valid only in Manual Mode and it is updated only after the command - update-***

7.3.3.5. **setR**

COMMAND
setR val
DESCRIPTION
Set the right manual blade position

*** This position is valid only in Manual Mode and it is updated only after the command - update-***

7.3.3.6. **setF**

setF val

DESCRIPTION

Set the front manual blade

*** This position is valid only if the collimator is in the open position

valid only if the collimator is in the open position
update-***

7.3.3.8. **setT**

COMMAND

setT val

DESCRIPTION

Set the Trap manual blade

*** This position is valid only if the collimator is in the open position
upd

update

COMMAND

update

DESCRIPTION

Update the collimator detector positions

7.3.4. 2D Filter activation

The following commands are used to activate the 2D filter

7.3.4.1. **selectFiltro**

This section

7.3.5.1. **getTomoFilterConf**

COMMAND

getTomoFilterConf

The following commands allow to control the Mirror position and the Light state.

7.3.6.1. **setMirror**

COMMAND
setMirror OUT/HOME
DESCRIPTION
Set the Mirror to one of the status: OUT or HOME

7.3.6.2. **setLamp**

COMMAND
setLamp ON/OFF
DESCRIPTION
Set the Lamp to one of the status: ON or OFF

7.3.6.3. **setCalibMirror**

COMMAND
setCalibMirror val
DESCRIPTION
Set the steps for the Mirror position when OUT

NOTE: this command changes only the memory not the configuration File. Use the command *STORE* to update the configuration File

7.3.7. Test commands

The following commands help to test the collimator device.

7.3.7.1. **setTrx**

COMMAND
setTrx angle
DESCRIPTION
Move the Tube to a target angle

7.3.7.2. **testColli**

COMMAND
testColli cycles

COMMAND

DESCRIPTION

Activate the collimation loop test for a given number of cycles

The Test performs sequential activation of the format blades, the filter, the Mirror.

NOTE: Avoid to activate a test with more of 50 cycles to prevent an overheat of the de

COMMAND

getCalibPo

The user shall measu^

- he true thickness at this compression;
- shall change the kF_val until the detected thickness equal the measured thickness.

7.4.4. Compression Force Parameters

7.4.4.1. *setWeight*

COMMAND
setWeight val
DESCRIPTION
Set the weigh of the Trolley

The weight is substracted from the detected compression force when the ARM is in CounterUpright position.

7.4.4.2. *setLimitForce*

COMMAND
setLimitForce val
DESCRIPTION
Set the maximum applicable compression force

The value can be set in a range from 70 to 200 Newton.

7.4.4.3. *setMinTarget*

COMMAND
setMinTarget val
DESCRIPTION

COMMAND

Set the minimum selectable Automatic target compression

The value can be set in the range: 30 to 200 Newton.

7.5. MOTOR ACTIVATION COMMANDS (rotazioni:)

7.5.1. Configuration File Management

7.5.1.1. *readTrxConfig*

COMMAND

readTrxConfig

D

Read the TRX configuration File

7.5.1.2. *saveTrxConfig*

COMMAND

saveTrxConfig

DESCRIPTION

Store the TRX configuration File

7.5.1.3. *readArmConfig*

COMMAND

readArmConfig

DESCRIPTION

Read the ARM configuration File

7.5.1.4. *saveArmConfig*

COMMAND

saveArmConfig

DESCRIPTION

Store the ARM configuration File

7.5.1.5. *readLenzeConfig*

COMMAND

DESCRIPTION

Activate the TRX to a target angle

The target can be of different nature:

- STOP: stops any current rotation;
- WHOME: move TRX to Wide Home position;
- NHOME: move TRX to Narrow Home position;
- IHOME: move TRX to Intermediate Home position;
- WEND: move TRX to Wide End position;
- NEND: move TRX to Narrow End position;
- IEND: move TRX to Intermediate End position;
- angle: the target is in (°). Example: TRX 10 moves TRX to 10°

7.5.3.2. **TRX LOOP**

COMMAND

TRX LOOP angle

DESCRIPTION

Activate the TRX in a loop mode

The TRX will move from +angle and -angle until a STOP command is provided.

7.5.3.3. **ARM**

COMMAND

ARM target

DESCRIPTION

Activate the ARM to a target angle

7.6. POTTER COMMANDS (potter:)

7.6.1. **setGrid2D**

COMMAND

setGrid2D ON/OFF

DESCRIPTION

Activate (ON) or Deactivate (OFF) the 2D grid (if present)

7.6.2. **setGrid3D**

COMMAND

setGrid3D ON/OFF

DESCRIPTION

Activate (ON) or Deactivate (OFF) the 3D grid (if present)

7.7. AWS S

werOff

COMMAND

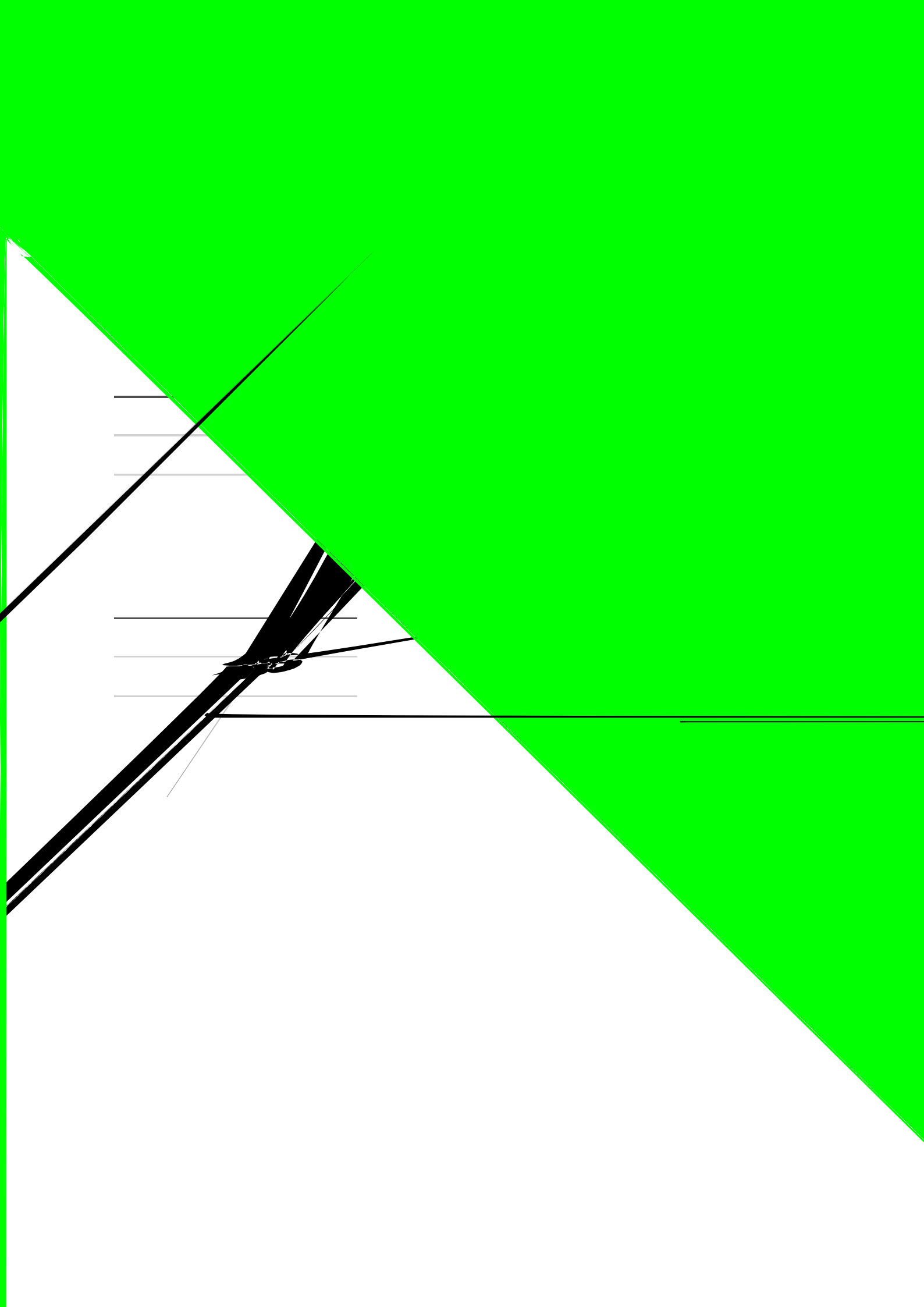
pcPowerOff frame

DESCRIPTION

Requests to the PC the permission to activate a Power Off sequence

The command can be used to test the

refer to the Biopsy device with the lateral Approach.



7.8.2.2. *calibSh*

COMMAND

calibSh zero +150_pos -150_pos

DESCRIPTION

Needle cursor

sensor position when the cursor is set to zero

sensor va

or value when the cursor is set to -15 mm;

COMMAND

DESCRIPTION

Activate the Buzzer for test purpose

7.8.3.5. **powerLed**

COMMAND

powerLed ON/OFF

DESCRIPTION

Activate (ON) or Deactivate (OFF) the Light

7.8.4. Configuration commands

7.8.4.1. **setYSensor**

COMMAND

setYSensor ON/OFF

DESCRIPTION

Activate (ON) or Deactivate (OFF) Y Upright position sensor

Note: Set this command OFF only for the old Biopsy devices that are not equipped with the Y Upright sensor

GENERATOR COMMANDS (generator:)

Position provided

- filament = *MoS* : select

ter:

- command = *HS* : activate the starter at the High Speed (if the High speed starter is present)

Exposure Overview

This section provides commands to open a special session that allow to activate a Manual Exposure for test purpose.

The Manual exposure allow to:

- Select to use a 35kv or 49 kV template file;
- Set the nominal value of the kV;
- Set the mAs of the pulse;
- Set the Focus to be used;
- Set the IDAC of the anodic current.

The Exposure has the following limitations:

- The maximum exposure time with kV < 35 is 100ms;
- The maximum exposure time with kV ≥ 35 is 50ms;
- The maximum IDAC is 2500;

In order to execute the Manual exposure:

- Open the session with the command: ***activateManualXray()***;
- Set the Exposure parameters to be used: ***setExposure()***;
- Press the X-Ray button to activate the procedure;
- To repeat the exposure with the same Exposure parameters only press again the X-RAY push button;
- When the session terminates, close the session with the command: ***closeManualXray()***

The Exposed data will be displayed at the sequence completion.

7.9.3.2. **activateManualXray**

COMMAND
activateManualXray tubeType
DESCRIPTION
Activate the Manual Exposure session uploading the Tube template file.
tubeType parameter:
• tubeType = "35KV": upload the TEMPLATE 35KV file; the E_XM1016THI configuratin file; and the Gantry is ready to expose.

SetNextExposure Focus kV mAs Idac
DESCRIPTION
Set the next exposure data
Focus parameter:
• Focus = "WL": will use the W large focus; • Focus = "WS": will use the W small focus; • Focus = "MoL": will use the Mo large focus; • Focus = "MoS": will use the Mo small focus;
kV parameter: set 20 to 49 kV (max 35 for 35 KV template file);
mAs parameter: set in a range 1 to 640 (it is time limited anyway);
Idac parameter:

COMMAND

Closes the manual Xray session

When the session is closed, the configured Tube configuration file is reloaded into the Gantry.

7.10. LOADER COMMANDS (loader:)

This section allows to upload a firmware to a remote Device.

The firmware that can be uploaded are:

- FW190.hex : the firmware of the PCB190 (generator);
- FW240.hex : the firmware of the PCB240 (service board);
- FW269.hex : the firmware of the PCB269 (compressor board);
- FW249U1.hex : the firmware of the PCB249 (collimator) Assy 2 controller U1;
- FW249U1A.hex : the firmware of the PCB249 Assy 1 controller U1;
- FW249U2.hex : the firmware of the PCB249 Assy 2 controller U2;
- FW249U2A.hex : the firmware of the PCB249 Assy 1 controller U1A (titer)

In order to upload the firmware to the target device, that firmware shall be properly for

related pre formatted firmware

B code:

the target shall be present into the /home/user directory of

DLS

help the Application debugging,
commands and methods.

7.11.1. DEVICE SERIAL PRO

COMMAND

Restart the execution of the Mater threads

NOTE: Only Expert. Dont't use this command!!

7.11.1.3. *read8*

COMMAND

read8 target address

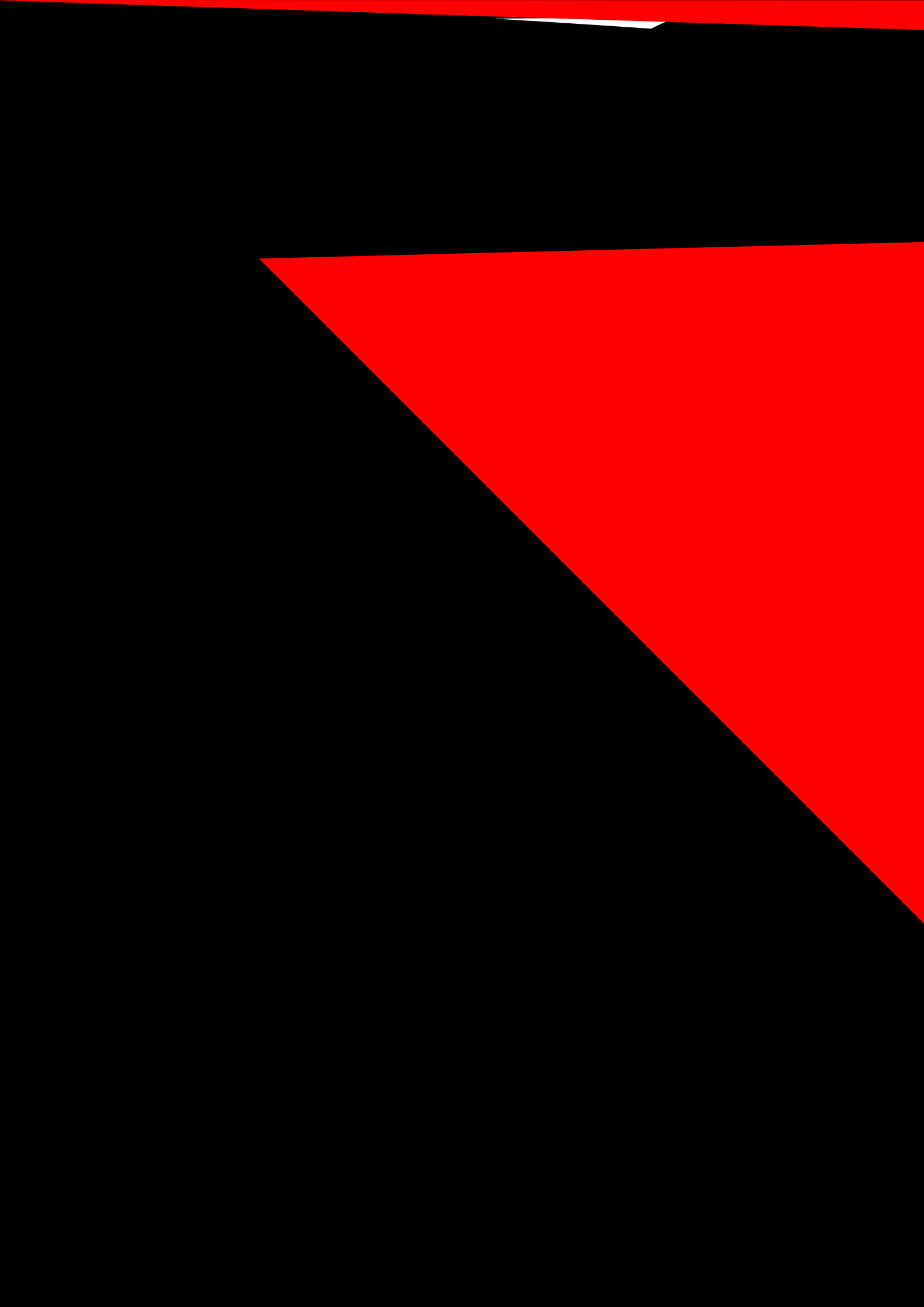
DESCRIPTION

Read the **address** 8 bit register from the **targ**

I

7.11.1.5. **write8**

COMMAND	
write8	target address value
DESCRIPTION	
Write the address 8 bit register of th	



the serial links the
man

Prin

and the serial links will be activated until
shown

2.7.2.7 Serial debug

The system has two separated Debug Terminals one for the Master activities
the Slave activities.

The Master terminal connection details are:

IP: 192.168.30.2;
PORT: 10010;
Protocol Raw;

The Slave terminal connection details are:

IP: 192.168.30.3,
PO